On Genetic Factors Influencing Educational Attainment

A Critical Essay on Heritability, Polygenic Scores, and Epigenetic Mechanisms

L. Rhodes (lcmrhodes98@gmail.com)
University College London

Produced as part of the Population Health Sciences (BSc)module "Frontiers of Knowledge in Population Health".

On Genetic Factors Influencing Educational Attainment

Introduction

This essay explores how genetics may influence educational attainment, and substantively evaluates the extent to which we can support the hypothesis that it does. First, a distinction is made between modern genetic research principals relative to social factors and how they differ from problematic perspectives of the early 20th century. Second, the importance of understanding the relationship between genetics and education is discussed. Following this, the concept of polygenic scores (*PGS*) and epigenetic clocks are defined and their potential use in social research to inform education policy is be explored. Finally, a conclusion is proposed considering an evidence-based approach to education policy that considers an individual's genome and epigenome, as well as social and environmental factors, through the inclusion of PGS and epigenetic clocks.

Distinguishing Between Eugenic Principals of the Early 20th Century and Modern Genetic Research

Throughout the early 20th century, eugenic thought permeated the study of genetics and its corresponding social implications. At University College London in 1911, the 'Galton Eugenics Professorship' was given to Karl Pearson, a foundational figure in modern statistics, and notably responsible for the development of Pearson's rho (the correlation coefficient). Pearson was against reforms relative to child labour and the minimum wage and suggested that such reforms encouraged reproduction among 'incapables' (Kevles, 1995). In the United States, similar contemporary proponents of the same period such as Harry Laughlin were fundamental in the construction of the notable 'Virginia Sterilization Act of 1924', wherein institutionalised individuals with a variety of hereditary cognitive disabilities were forcibly sterilised (Virginia General Assembly 1924). Thus, eugenics can be understood as categorising and prescribing preference to certain hereditary characteristics as a natural and desirable function of shaping society. In very stark contrast, much of the modern literature around hereditary factors (*including genetics by default*) takes the stance that they should be examined with the prospective goal of tailoring policies to disentangle the potential hereditary nature of any observed inequalities. Accordingly, where eugenicist thought might see associations between societal hierarchies and genetic factors as just and inevitable (Anderson, 1999), anti-eugenicist and egalitarian philosophy would seek to document how current systems perpetuate and reward certain genetic factors and then assess the extent to which both policy and future research may leverage these factors (Cesarini and Visscher, 2017; Harden, 2021). As such, it is the latter of these principles (*that of 'anti-eugenicist' thought*) that dictates the landscape of the discussed contemporary literature.

The Role of Genetics in Educational Attainment: The Twin Study / GWAS Gap

When considering genetics and educational attainment, we might first consider the results of contemporary twin studies. As such, utilising Swedish administrative data considering pairs of brothers born from 1951-1970, Cesarini and Visscher (2017) illustrate (*by use of Falconers formula* $H_b^2 = 2(r_{mz} - r_{dz})$) phenotypic heritability estimates of 58% for cognitive skills, comparative to 81% for a factor such as height. However, when adjusting for multiple sibling types and diminishing levels of phenotypic similarity, a proposed alternative estimation is 72% for cognitive skills and 94% for height. As cognitive skills may be considered as a primary influencing factor for educational attainment (Strenze, 2007), this evidence does provide support for the hypothesis that genetics influence educational attainment. Despite this, however, one cannot disregard the impact of environmental interventions upon such factors with purportedly high heritability. Just as we may consider that an individual with a genetic predisposition for a higher BMI or greater height may be impacted by their availability of requisite nutrition (Perkins et al, 2016), we may also consider the potential for similar gene-environment interactions relative to cognitive functioning, and by extension, educational attainment. However, rather than fully fledged 'genetically sensitive schools', as suggested by Asbury and Plomin (2013), Cesarini and Visscher only support the use of genetic information in the case that it supports current educational structures through effective improvements to interventions.

The contemporary literature on gene environment interactions regarding education seek to de-construct the nuances of the interaction. An example of such was observed recently regarding evidence for an interaction between school and students' educational attainment polygenic indices, including more than 23,000 families across Norway. Accordingly, it appeared that genetic effects (EA-PGI) were in fact weaker for higher performing schools - suggesting that either factors associated with teaching, behavioural factors such as enablement or the social modelling presented by higher performing schools was of increased relative importance (Cheesman et al, 2022). This indicates what whilst we do find support for the hypothesis that genetic factors influence educational attainment; genetic factors are unlikely to exist in isolation as the dominant predictor.

Whilst the use of PGIs (*polygenic index*) do provide a method for disseminating between genetic influence between factors, they also add for further complexity to understanding the extent of those factors' comparative to methods such as those employed in twin studies (Maher, 2008). Whilst purported levels of heritable explanation for cognitive function were seen to be as high as 74% in the previous examples, PGI's produced from samples ranging from 100,000-400,000 have only explained between 3-7% of the variation in a factor such as years of education (Cesarini, 2010; Rietveld, 2013; Okbay et al, 2016). Despite large sample sizes perceivably capturing more data and permitting increases in the amount of explainable estimated variation, the 'gap' between other methods (*such as twin studies*) and GWAS is likely due to the fact that GWAS are constructed according to current genotyping methods, and by extension, other genetic factors (*notably, epigenetic factors*) are not accounted for. In a twin study, these factors would be captured by default, being nested within the quantification of heritability.

The Role of Genetics in Educational Attainment: Dissecting a Recent Large-Scale GWAS

Further contemporary support for the hypothesis may be derived from a 2018 GWAS including data from 1.1 million European individuals, which led to the identification of 1,271 lead SNPs (*genome-wide significant independent SNPs*) when considering a significance threshold of P < 5 x 10ˆ-8 (Lee et al, 2018). Notably, 995 of which held at a stricter threshold of P < 1 × 10ˆ-8, with the median effect size of the lead SNPs (*genome-wide significant independent SNPs*) found being 1.7 weeks of schooling per allele (95% CI 1.1-2.6). As such, the resulting polygenic score produced from the study allowed for as much as 12% of the variance in educational attainment to be explained, however only when four separate phenotypes (educational attainment, cognitive performance, self-reported math ability, and hardest math class completed) were analysed together. Heterogeneity between effect sizes of SNP was also noted, especially regarding the within-family beach of the study.

Notably, in the within-family branch of the study, it was found that the genetic effects within sibling cohorts were 40% smaller than the results from a meta-analysis excluding siblings. As such, this indicates that GWAS effect sizes may be biased due to the correlation between educational attainment and a potentially positive educational environment in which an individual was raised in. Across cohorts, a genetic correlation of 0.72 was found, which was significantly different from 1 (p = 0.03). This further suggests that SNPs have different effects in different contextual settings.

To predict years of education by employing polygenic scores from the same study for cohorts from the National Longitudinal Study of Adolescent to Adult Health (*AddHealth*) and Health and Retirement Study (*HRS*), it was found that the additional variation explained by polygenic score (indicated by the incremental coefficient of determination) alongside other principal factors (*age, sex*) were 12.7% for AddHealth and 10.6% for HRS (*average 11.4%*). Comparative to other predictors, it was noted that polygenic score was a better predictor than household income, but not as good as the education level of an individual's parents. When demographic factors were included in addition to polygenic score, the incremental coefficient was found to drop to 4.6%, on average. That is, when holding constant contextual demographic factors, an approximate average 59.5% decrease in the amount of the variability in educational attainment explainable by the polygenic score was observed.

As the research conducts a meta-analysis of various cohorts and employs multiple sources of data, we might be satisfied that the level of the evidence exhibited by the research is good whilst attempting to disseminate the potential relationship between genetic factors and educational attainment. A considerable limitation, however, of the research is that it only considers individuals of European ancestry, thus leading to potential systematic bias if attempting to truly consider the phenotype relative to genetic factors. The research does attempt to account for this by drawing attention to this factor (*noting a considerably lower incremental coefficient for a cohort of African Americans from the HRS*), however it remains a limitation of the proposed polygenic scores and weakens future generalisability to other settings, such as research for policy relative to educational interventions.

Further, whilst educational attainment typically refers to the highest level of education completed by an individual (United States Census Bureau, 2023) the study employs a surrogate outcome of the simple number of years of

schooling completed by the individual. Whilst nuanced, this distinction indicates that the study may not have truly measured the phenotype in question. Consider the scenario in which an individual is made to repeat years of education due to repeated subpar academic achievement and finishes school at a later point than their peers. Whilst their quantitative years of schooling may be rather high, this does not necessarily mean they completed a high level of education (*as is the concept of educational attainment*). Preferentially, the study should explicitly define educational attainment conceptually in tandem with justification of the chosen method of measurement.

Overall, whilst the results of the study do find support for the hypothesis that genetic factors influence educational attainment in populations of European descent, we again see evidence that other contextual factors, and potential gene environment interactions indicated by heterogeneity of effect are (*at least*) equally as important in explaining and predicting educational attainment within this population. This indicates that the implied combinatory inclusion of genetic and contextual factors may be an important future approach for research and policy.

The Intersection of Genetics and Environmental Factors

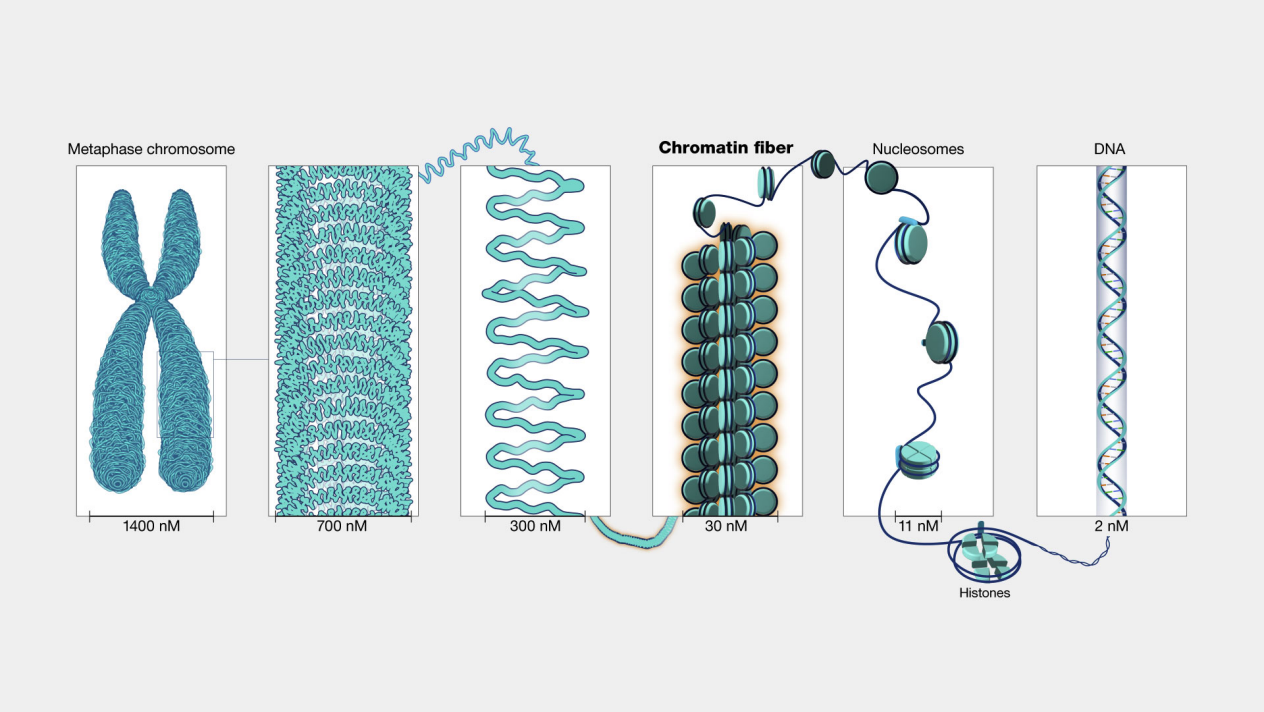

Figure 1: Chromosome Composition (Liu, 2023)

An important contemporary component pertinent to the hypothesis [that genetic factors influence educational attainment] is likely contained within the emerging science of epigenetics. That is, alterations in genome function, such as gene expression, that "cannot be explained by changes in DNA sequence" (Riggs et al, 1996). Accordingly, whilst all cells contain the same DNA, they do not all exhibit the same function. One component of this differing function is how the DNA is expressed within a cell. It is this process that is referred to as 'epigenetic modification' (Handy et al, 2011). Within a chromosome, DNA is stored within chromatin, and wrapped around nucleosomes which

are composed of histone proteins (see figure 1). One epigenetic mechanism is that of DNA methylation (**DNAm**), wherein a methyl group is attached directly to a cytosine residue within a cytosine-guanine (**CpG**) sequence, typically resulting in gene silencing (Moore, et al, 2013). Another epigenetic mechanism is histone modification, wherein acetyl or methyl groups bind to the histone 'tails', which modifies gene expression through altering the accessibility of parts of the gene for activation (Bannister and Kouzarides, 2011).

Typically, DNAm is reset during gametogenesis and early embryogenesis, and new DNAm patterns are established during implantation (Cedar and Bergman, 2012). Principally, DNAm is necessary for processes such as transcriptional regulation in embryonic development and functions of gene expression (Jjingo et al, 2012). According to the literature, DNAm (*and other epigenetic mechanisms*) are influenced both by genes themselves, various diseases throughout life, and also exposure to compounds such as Bisphenol A (BPA), lead and mercury (Hanna et al, 2012; Martin and Fry, 2018). Environmental factors such as tobacco smoke exposure, exposure to various micronutrients and vitamins and even socioeconomic status and maternal depressive disorders have also been linked to altered DNA methylation (Borghol et al, 2012; Braithwaite et al, 2015; Mansell et al, 2016).

Whilst various factors have been found to influence the epigenome throughout an individual's life, a variety of hypothesis exist for how they may also operate in a potential inter-generational manner through various impacts occurring in gestation or relative to the intrauterine environment. For instance, through potential regulation of gene expression by ncRNAs at transcriptional and translational levels, and though the potential transmission of ncRNAs via sperm or maternal germ cells establishing epigenetic foundations (Lacal and Ventura, 2018). Thus, whilst Mendel's laws of inheritance and the Boveri-Sutton chromosome theory dictate that only an individual's chromosomes and associated germline mutations may be inherited, and that environmental adjustments or somatic mutations may not be inherited remain the dominant paradigm in the field of biological inheritance, advances in epigenetics may confound the nuances of this assumption (O'Connor and Miko, 2008). Accordingly, such factors may have the potential to stimulate a converse resurgence of 'Lamarckian' thought.

Epigenetics and Educational Attainment

Epigenetics may present themselves as one of the key 'missing links' between the aforementioned discrepancy between twin studies and the results of GWAS regarding educational attainment. Recent research has suggested that greater collective epigenetic indications of aging as modelled by 'epigenetic clocks' may be associated with decreased educational attainment. As we are aware that DNA methylation and other epigenetic mechanisms may be influenced by a variety of environmental factors and exposures, the potential linkage of these factors to the phenotype of educational attainment would even further support the hypothesis that genetic factors influence educational attainment, through providing a direct link between environments and genetic information itself.

This was posited by a recent study that examined multiple posited 'epigenetic' clocks of aging and found various associations with lower educational attainment (Crimmins et al, 2021). In 6 of the 13 considered clocks, decreased years of education were significantly associated with increased epigenetic indications of ageing, whilst holding constant other variables such as race, bmi, smoking status and gender. Where results for education were significant, regression coefficients were observed to increase as years of schooling decreased, with the GrimAge clock indicating

that 0-11 years of education were associated aging just over two years faster (2.03) than the reference group (*of 16+ years of education*). It is notable that all the considered 'second generation' clocks (*Zhang - AccelAge, Levine - PhenoAccelAge, GrimAge - AccelAge, and DunedinPoAm38 - AccelAge*) - wherein health indicators were included in the training of the clocks in addition to age alone - found the correlations with decreased educational attainment statistically significant, when considering a significance threshold of 5%. This independently indicates that more complete training and advances in the complexity of such measures may facilitate even greater potential links between the environment and the epigenome.

However, while some of these results do certainly indicate various linear degrees of association with the considered phenotype, various limitations exist regarding these results. Firstly, for 7 of the 13 clocks considered, significant results for educational attainment were not found. As previously mentioned, this suggests that first generation epigenetic clocks may not capture important nuances with social phenotypes such as educational attainment, and thus dictates necessary caution in their application for future use cases. In addition, whilst the data used by the study is nationally representative of Americans over 50, this limits the extent to which we can generalise these results in other instances.

Accordingly, epigenetic clocks could be an important tool for research. They could be applied in generalised regression techniques in social research as a confounding independent variable, in addition to tools such as polygenic scores. However, as 'second generation' clocks appear more sensitive to social phenotypes, explicit highlighting and consideration would have to be followed through in their application, as they may be sensitive to further developments in the field and results be later rendered inaccurate.

In addition, the fact we cannot generalise these results beyond Americans over 50 is a considerable limitation. If we consider that changes to the epigenome are related to an individual's environment, it is not unreasonable to assume that the environment experienced by this population subgroup will be very different to those in other countries, and even later generations of Americans. That is to say, the educational environment experienced by Americans over 50 is likely to be very different from the educational environment of Americans (and other nationalities) today. Thus, whilst we may find the results from second generation epigenetic clocks promising in relation to educational attainment, further research considering a more diverse study population would assist in distilling appropriate further use. In addition, future studies could consider a prospective cohort study design to ensure that potential forms of survivorship bias do not confound any observed correlations.

Conclusion

In conclusion, we have found a variety of evidence that supports the hypothesis that genetics likely influence educational attainment. Our research postulates that the use of PGS (*polygenic scores*) and epigenetic clocks in future social research could form the basis for evidence-based education policy, particularly in addressing socioeconomic inequalities. As a recent report by the Education Policy Institute highlights, approximately £250 million is spent on widening participation for disadvantaged students (Robinson and Salvestrini, 2020). However, there is inconclusive evidence regarding the effectiveness of these interventions on enrolment.

By controlling for PGS or epigenetic clock data in evaluations, greater precision in understanding the effectiveness of such policies can be achieved. This approach would provide a more comprehensive understanding of an individual's interaction with education, taking into account both the genome and epigenome in addition to social and environmental factors. This evidence-based approach would shift focus away from the concept of "genetically sensitive schools" and toward creating educational policies centred around accurate evidence-based research and "proportionate universalism" within the education sector - that is, the universal provision of services proportionate to the specific dynamics of need (Marmot and Bell, 2012).

References

Anderson, E. S. (1999). What is the point of equality? Ethics, 109(2), 287-337. Available at: https://doi.org/10.1086/233897 (Accessed 18th January 2023).

Asbury, K. and Plomin, R. (2013). G is for Genes: The Impact of Genetics on Education and Achievement. Wiley-Blackwell.

Bannister, A. J., and Kouzarides, T. (2011). Regulation of chromatin by histone modifications. Cell Research, 21 (3), 381-395. Available at: https://doi.org/10.1038/cr.2011.22 (Accessed 21st January 2023)

Borghol, N., Suderman, M., McArdle, W., Racine, A., Hallett, M., Pembrey, M., Hertzman, C., Power, C., and Szyf, M. (2012). Associations with early-life socio-economic position in adult DNA methylation. International Journal of Epidemiology, 41 (1), 62-74. Available at: https://doi.org/10.1093/ije/dyr147 (Accessed 23rd January 2023)

Braithwaite, E. C., Kundakovic, M., Ramchandani, P. G., Murphy, S. E., and Champagne, F. A. (2015). Maternal prenatal depressive symptoms predict infant NR3C1 1F and BDNF IV DNA methylation. Epigenetics, 10 (5), 408-417. Available at: https://doi.org/10.1080/15592294.2015.1039221 (Accessed 23rd January 2023)

Cedar, H., and Bergman, Y. (2012). Programming of DNA Methylation Patterns. Annual Review of Biochemistry, 81(1), 97-117. Available at: https://doi.org/10.1146/annurev-biochem-052610-091920 (Accessed 21st January 2023)

Cesarini, D. (2010) Essays on Genetic Variation and Economic Behavior. Massachusetts Institute of Technology.

Cesarini, D. and Visscher, P. M. (2017). Genetics and Educational attainment.

Cheesman, R., Borgen, N. T., Lyngstad, T. H., Eilertsen, E. M., Ayorech, Z., Torvik, F. A., Andreassen, O. A., Zachrisson, H. D., and Ystrom, E. (2022). A population-wide gene-environment interaction study on how genes, schools, and residential areas shape achievement. npj Science of Learning, 7 (1), 29. Available at: https://doi.org/10.1038/s41539-022-00145-8 (Accessed January 19th 2023)

Crimmins, E. M., Thyagarajan, B., Levine, M. E., Weir, D. R., and Faul, J. (2021). Associations of Age, Sex, Race/Ethnicity, and Education With 13 Epigenetic Clocks in a Nationally Representative U.S. Sample: The Health and Retirement Study. The Journals of Gerontology: Series A, 76(6), 1117-1123. Available at: https://doi.org/10.1093/gerona/glab016 (Accessed 26th January 2023)

Handy, D. E., Castro, R., and Loscalzo, J. (2011). Epigenetic modifications: basic mechanisms and role in cardiovascular disease. Circulation, 123(19), 2145–2156. Available at: https://doi.org/10.1161/CIRCULATIONAHA.110.956839 (Accessed 21st January 2023)

Hanna, C. W., Bloom, M. S., Robinson, W. P., Kim, D., Parsons, P. J., vom Saal, F. S., Taylor, J. A., Steuerwald, A. J., and Fujimoto, V. Y. (2012). DNA methylation changes in whole blood is associated with exposure to the environmental contaminants, mercury, lead, cadmium and bisphenol A, in women undergoing ovarian stimulation

for IVF. Human Reproduction, 27(5), 1401-1410. Available at: https://doi.org/10.1093/humrep/des038 (Accessed 23rd January 2023)

Harden, K. P. (2021). The Genetic Lottery: Why DNA Matters for Social Equality. Princeton University Press.

Jjingo, D., Conley, A. B., Yi ,S. V., Lunyak, V. V., Jordan, I. (2012) On the presence and role of human gene-body DNA methylation. Oncotarget; 3: 462-474. Available at: https://www.oncotarget.com/article/497/text/ (Accessed 23rd January 2023)

Kevles, D. J. (1995). In the Name of Eugenics: Genetics and the Uses of Human Heredity. Harvard University Press.

Lacal, I., and Ventura, R. (2018). Epigenetic Inheritance: Concepts, Mechanisms and Perspectives [Review]. Frontiers in Molecular Neuroscience, 11. Available at: https://doi.org/10.3389/fnmol.2018.00292 (Accessed 23rd January 2023)

Lee, J. J., Wedow, R., Okbay, A., Kong, E., Maghzian, O., Zacher, M., Nguyen-Viet, T. A., Bowers, P., Sidorenko, J., Karlsson Linnér, R., Fontana, M. A., Kundu, T., Lee, C., Li, H., Li, R., Royer, R., Timshel, P. N., Walters, R. K., Willoughby, E. A., . . . Social Science Genetic Association, C. (2018). Gene discovery and polygenic prediction from a genome-wide association study of educational attainment in 1.1 million individuals. Nature Genetics, 50 (8), 1112-1121. Available at: https://doi.org/10.1038/s41588-018-0147-3 (Accessed 20th January 2023)

Liu, P. P. (2023) Chromatin. [JPG] Available at: https://www.genome.gov/genetics-glossary/Chromatin (Accessed 3rd February 2023)

Maher, B. (2008) The case of the missing heritability. Nature 456, 18–21. [Online] Available at: https://go.gale.com/ps/i.do?id=GALE%7CA189000843&sid=googleScholar&v=2.1&it=r&linkaccess=abs&issn=00280836&p=HRCA&sw=w&userGroupName=anon%7Ea57c8ac1 (Accessed January 20th 2023)

Mansell, T., Novakovic, B., Meyer, B., Rzehak, P., Vuillermin, P., Ponsonby, A. L., Collier, F., Burgner, D., Saffery, R., Ryan, J., Vuillermin, P., Ponsonby, A.-L., Carlin, J. B., Allen, K. J., Tang, M. L., Saffery, R., Ranganathan, S., Burgner, D., Dwyer, T., . . . team, B. I. S. i. (2016). The effects of maternal anxiety during pregnancy on IGF2/H19 methylation in cord blood. Translational Psychiatry, 6 (3), e765-e765. Available at: https://doi.org/10.1038/tp.2016.32 (Accessed 23rd January 2023)

Marmot, M., and Bell, R. (2012). Fair society, healthy lives. Public health, 126 Suppl 1, S4–S10. Available at: https://doi.org/10.1016/j.puhe.2012.05.014 (Accessed 28th January 2023)

Martin, E. M., and Fry, R. C. (2018). Environmental Influences on the Epigenome: Exposure- Associated DNA Methylation in Human Populations. Annual Review of Public Health, 39(1), 309-333. Available at: https://doi.org/10.1146/annurev-publhealth-040617-014629 (Accessed 23rd January 2023)

Moore, L. D., Le, T., and Fan, G. (2013). DNA Methylation and Its Basic Function. Neuropsychopharmacology, 38 (1), 23-38. Available at: https://doi.org/10.1038/npp.2012.112 (Accessed 21st January 2023)

npj Science of Learning, 2 (1), 4. Available at: https://doi.org/10.1038/s41539017-0005-6 (Accessed 18th January 2023) O'Connor, C. and Miko, I. (2008) Developing the chromosome theory. Nature Education 1 (1):44. Available at: https://www.nature.com/scitable/topicpage/developing-the-chromosome-theory-164/ (Accessed 24th January 2023)

Okbay, A., Beauchamp, J. P., Fontana, M. A., Lee, J. J., Pers, T. H., Rietveld, C. A., Turley, P., Chen, G.-B., Emilsson, V., Meddens, S. F. W., Oskarsson, S., Pickrell, J. K., Thom, K., Timshel, P., de Vlaming, R., Abdellaoui, A., Ahluwalia, T. S., Bacelis, J., Baumbach, C., . . . Benjamin, D. J. (2016). Genome-wide association study identifies 74 loci associated with educational attainment. Nature, 533 (7604), 539-542. Available at: https://doi.org/10.1038/nature17671 (Accessed 20th January 2023)

Perkins, J. M., Subramanian, S. V., Davey Smith, G., and Özaltin, E. (2016). Adult height, nutrition, and population health. Nutrition Reviews, 74 (3), 149-165. Available at: https://doi.org/10.1093/nutrit/nuv105 (Accessed January 19th 2023) Rietveld, C. A., Medland, S. E., Derringer, J., Yang, J., Esko, T., Martin, N. W., Westra, H.-J., Shakhbazov, K., Abdellaoui, A., Agrawal, A., Albrecht, E., Alizadeh, B. Z., Amin, N., Barnard, J., Baumeister, S. E., Benke, K. S., Bielak, L. F., Boatman, J. A., Boyle, P. A., . . . Koellinger, P. D. (2013). GWAS of 126,559 Individuals Identifies Genetic Variants Associated with Educational Attainment. Science, 340 (6139), 1467-1471. Available at: https://doi.org/10.1126/science.1235488 (Accessed 20th January 2023)

Riggs, A. D., Martienssen, R. A., and Russo, V. E. (1996). Introduction. In V. E. A. Russo, R. A. Martienssen, and A. D. Riggs (Eds.), Epigenetic mechanisms of gene regulation. Plainview, NY: Cold Spring Harbor Laboratory Press.

Robinson, D. and Salvestrini, V. (2020). The Impact of Interventions for Widening Access to Higher Education: A Review of the Evidence. [Online] Available at: https://epi.org.uk/wp-content/uploads/2020/03/Widening_participation-review_EPI-TASO_2020-1.pdf (Accessed 26th January 2023)

Strenze, T. (2007). Intelligence and socioeconomic success: A meta-analytic review of longitudinal research. Intelligence, 35 (5), 401-426. Available at: https://doi.org/https://doi.org/10.1016/j.intell.2006.09.004 (Accessed January 19th 2023)

Virginia General Assembly. (1924). An act to provide for the sexual sterilization of inmates of State institutions in certain cases. Chap. 394. [S B 281].